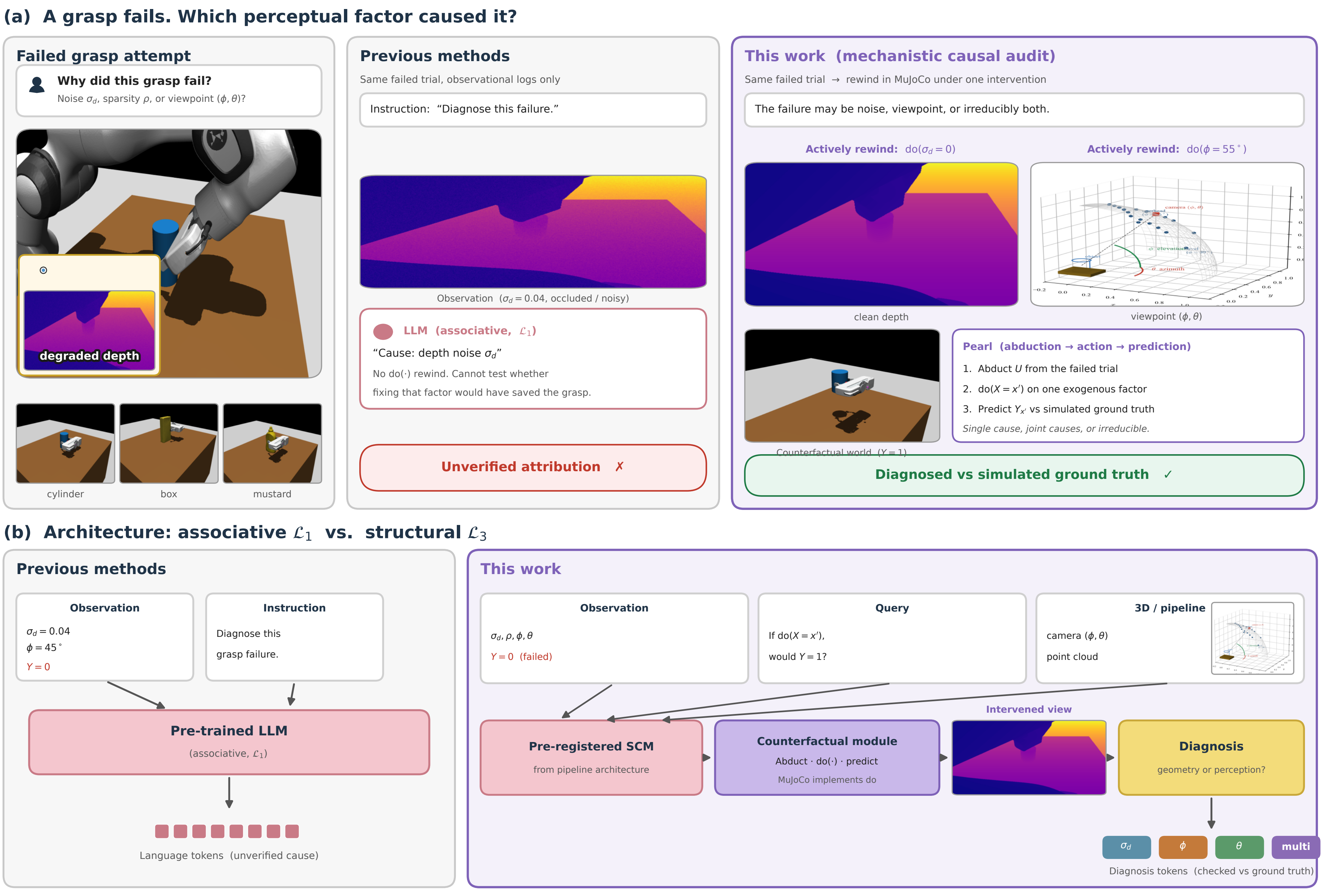

(a) A grasp fails. Which perceptual factor caused it?
Failed grasp attempt
Why did this grasp fail?
Noise σ_d, sparsity ρ, or viewpoint (ϕ, θ)?
degraded depth
cylinder
box
mustard
Previous methods
Same failed trial, observational logs only
Instruction: "Diagnose this failure."
Observation ($\sigma_d = 0.04$, occluded / noisy)
LLM (associative, $\mathcal{L}_1$)
"Cause: depth noise σ_d"
No do(·) rewind. Cannot test whether
fixing that factor would have saved the grasp.
Unverified attribution ✗
This work (mechanistic causal audit)
Same failed trial → rewind in MuJoCo under one intervention
The failure may be noise, viewpoint, or irreducibly both.
Actively rewind: do($\sigma_d = 0$)
Actively rewind: do($\phi = 55^\circ$)
clean depth
viewpoint (ϕ, θ)
Counterfactual world ($Y = 1$)
Pearl (abduction → action → prediction)
1. Abduct U from the failed trial
2. do($X = x'$) on one exogenous factor
3. Predict $Y_{x'}$ vs simulated ground truth
Single cause, joint causes, or irreducible.
Diagnosed vs simulated ground truth ✓
(b) Architecture: associative $\mathcal{L}_1$ vs. structural $\mathcal{L}_3$
Previous methods
Observation
$\sigma_d = 0.04$
$\phi = 45^\circ$
$Y = 0$
Instruction
Diagnose this
grasp failure.
Pre-trained LLM
(associative, $\mathcal{L}_1$)
Language tokens (unverified cause)
This work
Observation
$\sigma_d, \rho, \phi, \theta$
$Y = 0$ (failed)
Query
If do($X = x'$),
would $Y = 1$?
3D / pipeline
camera (ϕ, θ)
point cloud
Intervened view
Pre-registered SCM
from pipeline architecture
Counterfactual module
Abduct · do(·) · predict
MuJoCo implements do
Diagnosis
geometry or perception?
σ_d
ϕ
θ
multi
Diagnosis tokens (checked vs ground truth)